

4. Hashtag popularity analysis: Determine which hashtags are most popular among different user groups.

```
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter
import re
```

```
df = pd.read_csv('location.csv')
df.head()
```

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	H
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	
2	2	2	Just finished an amazing workout! 👏 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	
3	3	3	Excited about the upcoming weekend	Positive	2023-01-15	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	

Next steps:

[Generate code with df](#)[View recommended plots](#)[New interactive sheet](#)

```
tags = []
for text in df['Hashtags']:
    tags.extend(re.findall(r'[a-z|A-Z]*', text))
print(tags)
```

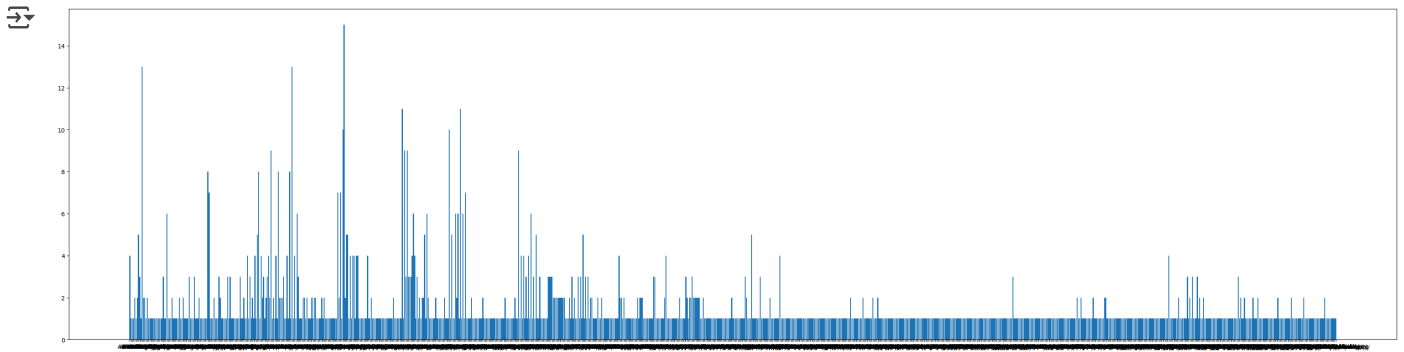
```
['#Nature', '#Park', '#Traffic', '#Morning', '#Fitness', '#Workout', '#Travel', '#Adventure', '#Cooking', '#Food', '#Gratitude', '#F
```

```
element_count = Counter(tags)
print(element_count)
```

```
Counter({'#Serenity': 15, '#Gratitude': 13, '#Excitement': 13, '#Despair': 11, '#Nostalgia': 11, '#Contentment': 10, '#Curiosity': 1
```

```
counts = []
elements = []
for x in element_count:
    counts.append(element_count.get(x))
    elements.append(x)
```

```
plt.figure(figsize=(40,10))
plt.bar(elements, counts)
plt.show()
```



```
top_hashtags = element_count.most_common(10)
tags, counts = zip(*top_hashtags)
```

```
plt.figure(figsize=(12,6))
plt.bar(tags, counts, color='skyblue')
plt.title("Top 10 Hashtags")
plt.xlabel("Hashtags")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

